Physics Sure Shot Questions: Chapter-wise Analysis

Comprehensive Master List of High-Priority Topics & Derivations

Chapter 01: Electric Charges and Fields

- **Derivations:** Derive the expression for the electric field due to an electric dipole at a point on its **axial line** and on its **equatorial line**.
- **Gauss's Law:** State Gauss's law and use it to find the electric field due to an infinitely long straight uniformly charged wire or a uniformly charged infinite plane sheet.
- **Flux & Superposition:** Calculate electric flux through shapes in a uniform electric field, and solve problems involving forces between multiple point charges.

Chapter 02: Electrostatic Potential and Capacitance

- **Capacitors:** Derive the expression for the capacitance of a parallel plate capacitor with a dielectric slab inserted between the plates.
- **Equipotential Surfaces:** Define equipotential surfaces and draw them for point charges or uniform fields. Explain why field lines are perpendicular to these surfaces.
- **Energy:** Derive the energy stored in a capacitor and potential energy of a system of charges.

Chapter 03: Current Electricity

- **Drift Velocity:** Define drift velocity and relaxation time. Derive the relationship between electric current and drift velocity, and deduce Ohm's law or resistivity.
- **Kirchhoff's Rules:** Use them to solve circuit problems or to derive the balance condition for a **Wheatstone bridge**.

- **Cells:** Internal resistance, emf, and terminal voltage calculations for series and parallel combinations.

Chapter 04: Moving Charges and Magnetism

- **Biot-Savart Law:** Derive the magnetic field on the axis of a circular current-carrying loop.
- **Forces:** Derive the force between two parallel current-carrying conductors and define the **ampere**.
- **Galvanometer:** Explain the principle and working; conversion into an ammeter or voltmeter.

Chapter 05: Magnetism and Matter

- **Magnetic Properties:** Distinguish between **diamagnetic, paramagnetic, and ferromagnetic** substances based on susceptibility and behavior in fields.
- **Bar Magnets:** Properties of magnetic field lines and comparison of a solenoid to a bar magnet.

Chapter 06: Electromagnetic Induction

- **Inductance:** Define mutual and self-inductance. Derive mutual inductance of two coaxial solenoids.
- **Lenz's Law:** State the law and show compliance with the conservation of energy.
- **AC Generator:** Working principle and derivation of the expression for induced emf.

Chapter 07: Alternating Current

- **LCR Circuit:** Derive the impedance and explain the condition for **resonance**.
- **Transformer:** Working principle, types of losses, and voltage/current relationships.
- **Power:** Average power in an AC circuit and definition of the **power factor**.

Chapter 08: Electromagnetic Waves

- **EM Spectrum:** Frequency ranges and applications for radar, water purification (UV), etc.
- **Properties:** Transverse nature, speed relations, and the concept of **displacement current**.

Chapter 09: Ray Optics and Optical Instruments

- **Lenses & Prisms:** Derive the **Lens Maker's Formula** and the refractive index relation for a **prism**.
- **Instruments:** Ray diagrams for astronomers' telescope and compound microscope (with magnifying power).
- **Total Internal Reflection:** Conditions and applications like optical fibers.

Chapter 10: Wave Optics

- **Huygens' Principle:** Define wavefront and verify laws of **reflection** or **refraction**.
- **Interference:** Fringe width derivation for **Young's Double Slit Experiment** (YDSE).
- **Diffraction:** Single slit diffraction; secondary maxima/minima conditions.

Chapter 11: Dual Nature of Radiation and Matter

- **Photoelectric Effect:** Emission laws and **Einstein's photoelectric equation** (threshold frequency/stopping potential).
- **Matter Waves:** Calculation of **de Broglie wavelength** for accelerated particles.
- **Graphs:** Photoelectric current vs. intensity, potential, and frequency interpretation.

Chapter 12: Atoms

- **Bohr Model:** Derive **radius** and **total energy** of the nth orbit of a hydrogen atom.

- **Spectrum:** Explain spectral series (Lyman, Balmer, etc.) using the Rydberg formula.

Chapter 13: Nuclei

- **Binding Energy:** BE per nucleon plot vs. mass number; nuclear stability, fission, and fusion conclusions.
- **Density:** Prove nuclear density is independent of mass number A.
- **Fission & Fusion:** Distinctions and energy release calculations.

Chapter 14: Semiconductor Electronics

- **PN Junction:** Depletion region formation and V-I characteristics for forward/ reverse bias.
- **Rectifiers:** Working of **half-wave** and **full-wave rectifiers**.
- **Band Theory:** Band diagrams for conductors, semiconductors, and insulators.

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